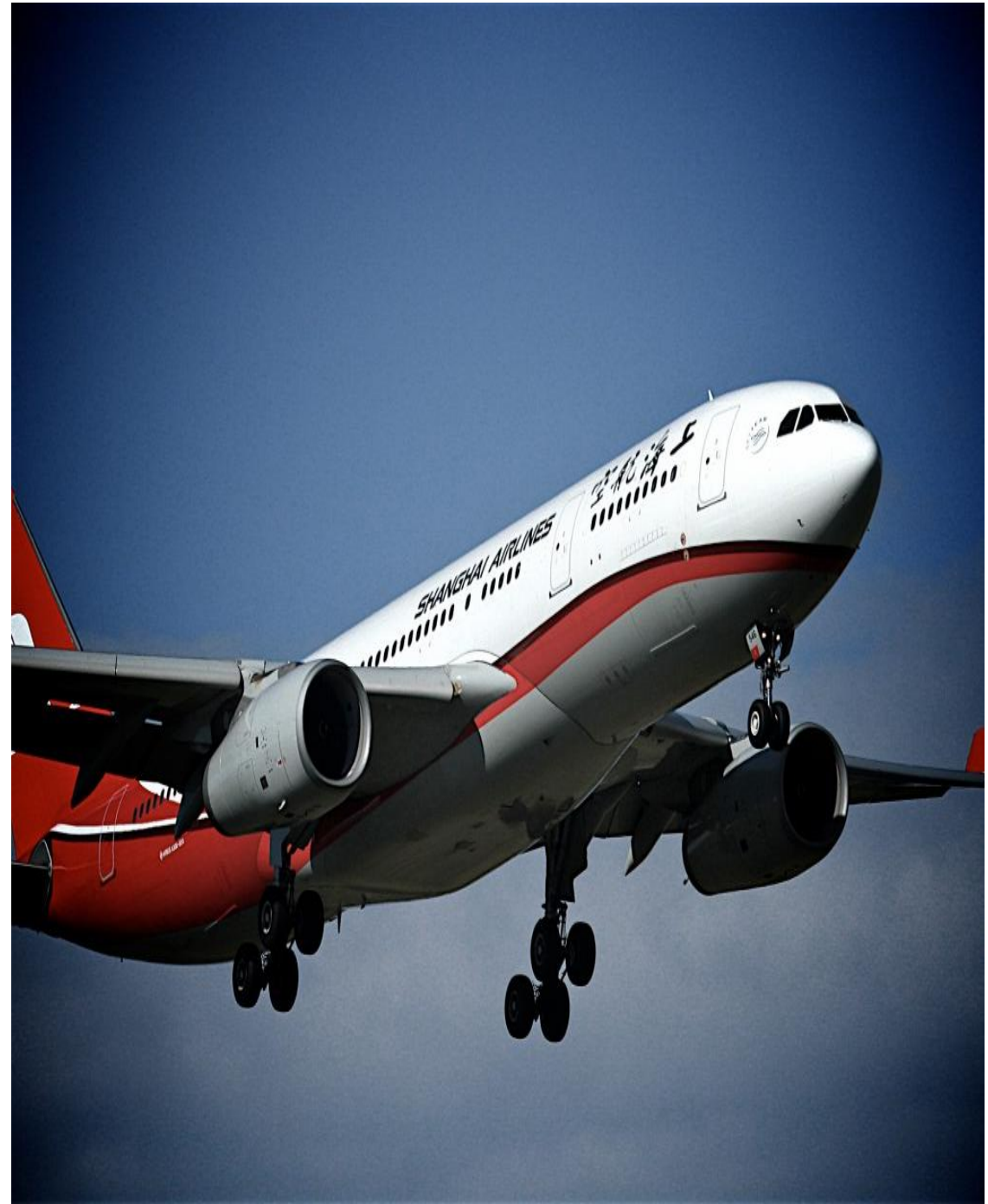


OPERATIONAL RISK ANALYSIS FOR AIRCRAFT ACQUISITION



Business overview

- The company wants to delve into the Aviation industry.
- There are several impending operational risks which demand careful selection of aircraft.

Goal

- ☐ To determine the aircraft with the lowest operational risk for private and commercial operations

Key Questions

- Which airplane has the least operational risk overall?
- What is the effect of weather conditions on airplane operations?
- Which aircraft make shows the lowest rate of fatal injuries?

Data overview

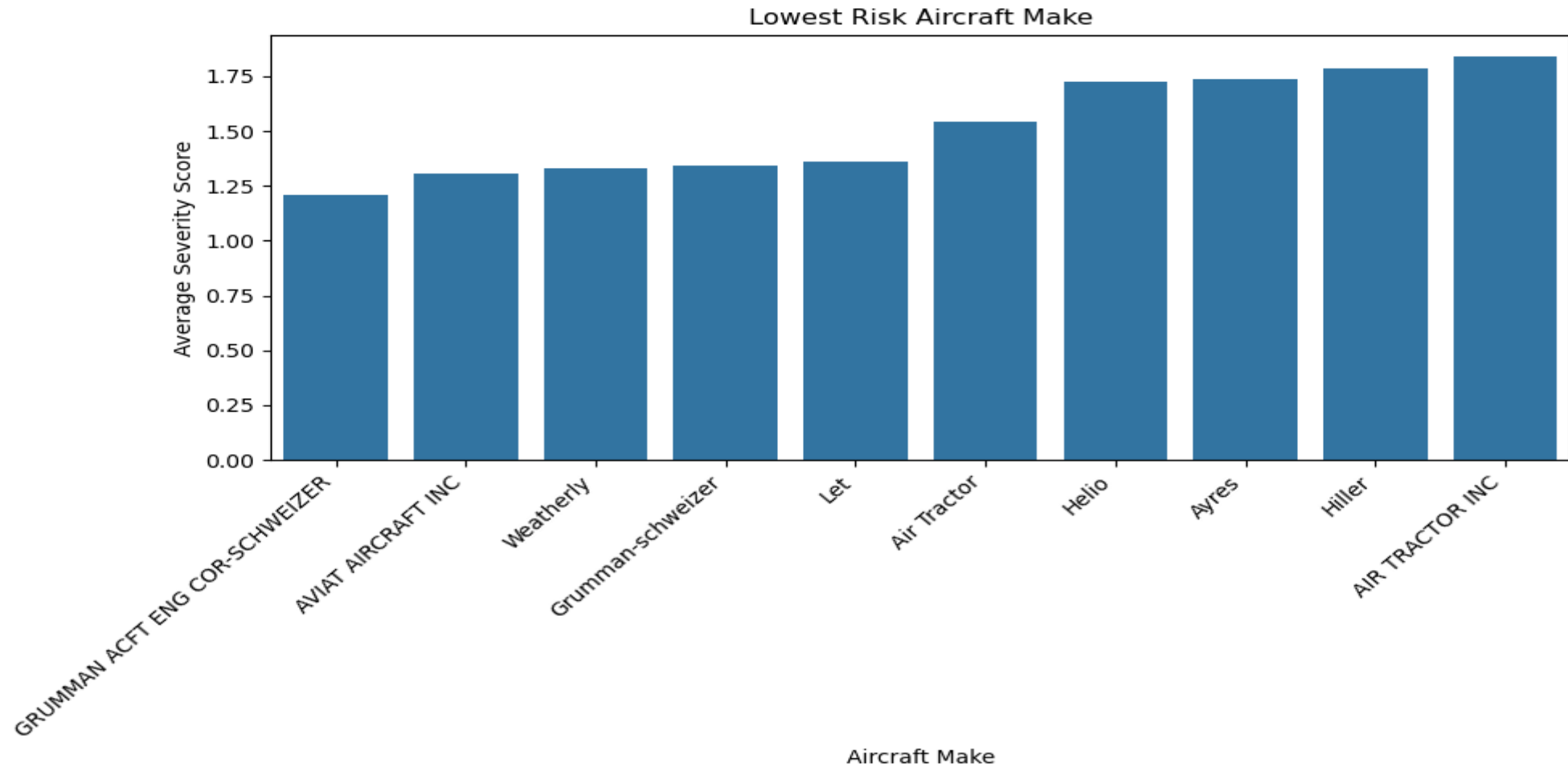
- The dataset to be used is publicly available and consists historical accident data.
- It contains 90348 recorded accidents and 31 variables including weather conditions, aircraft make and model and injuries among others

Risk calculation

- A weighted severity score was calculated as follows:
- $\text{Weighted severity score} = \text{category} * \text{weight}$

Category	Weight
Total Fatal Injuries	5
Total Serious Injuries	4
Total Minor Injuries	3

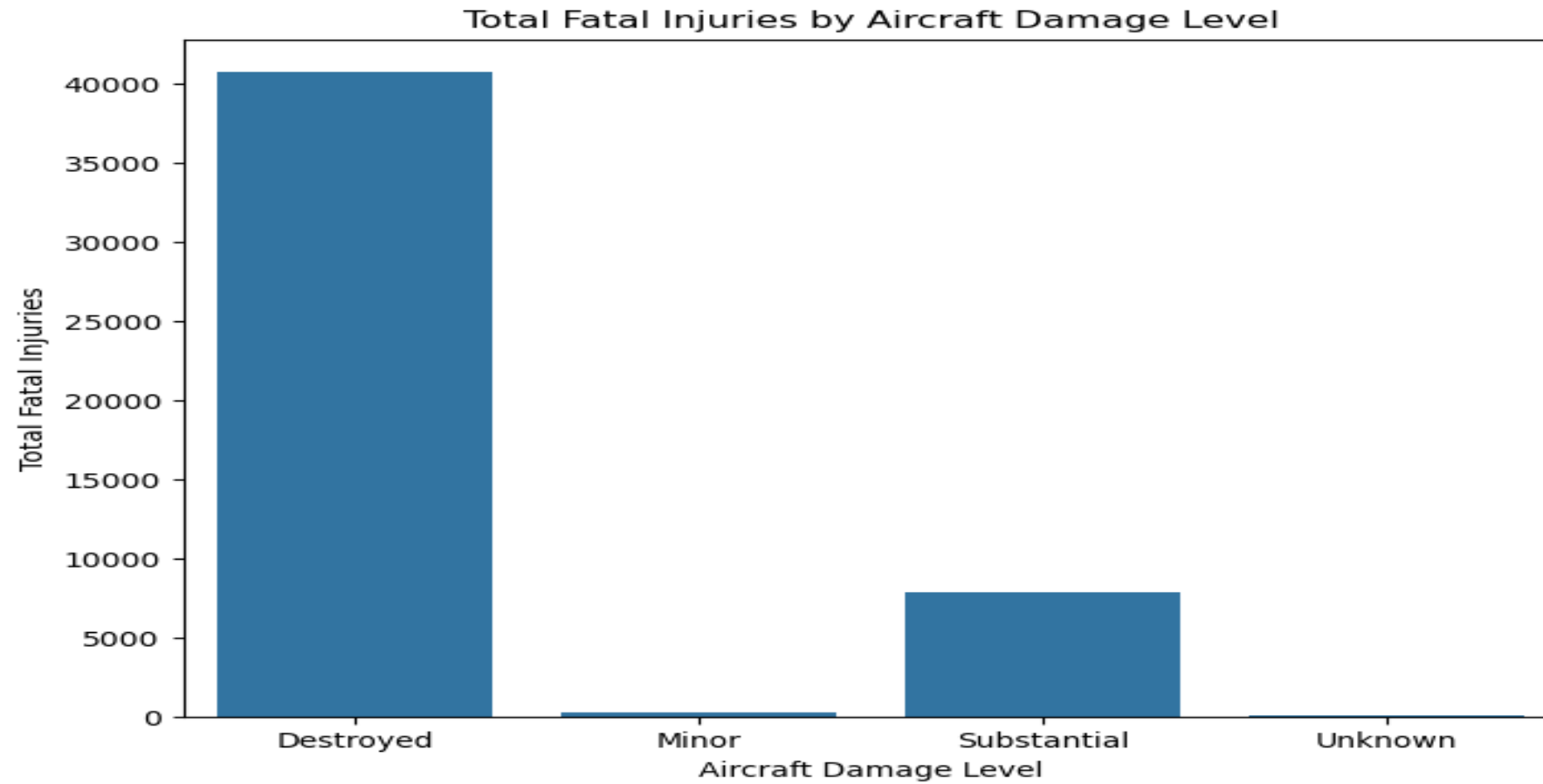
Visualization 1



Finding(viz 1)

- Certain aircraft make are associated with low accident severity score.
- This makes them safer and suitable for aircraft operations

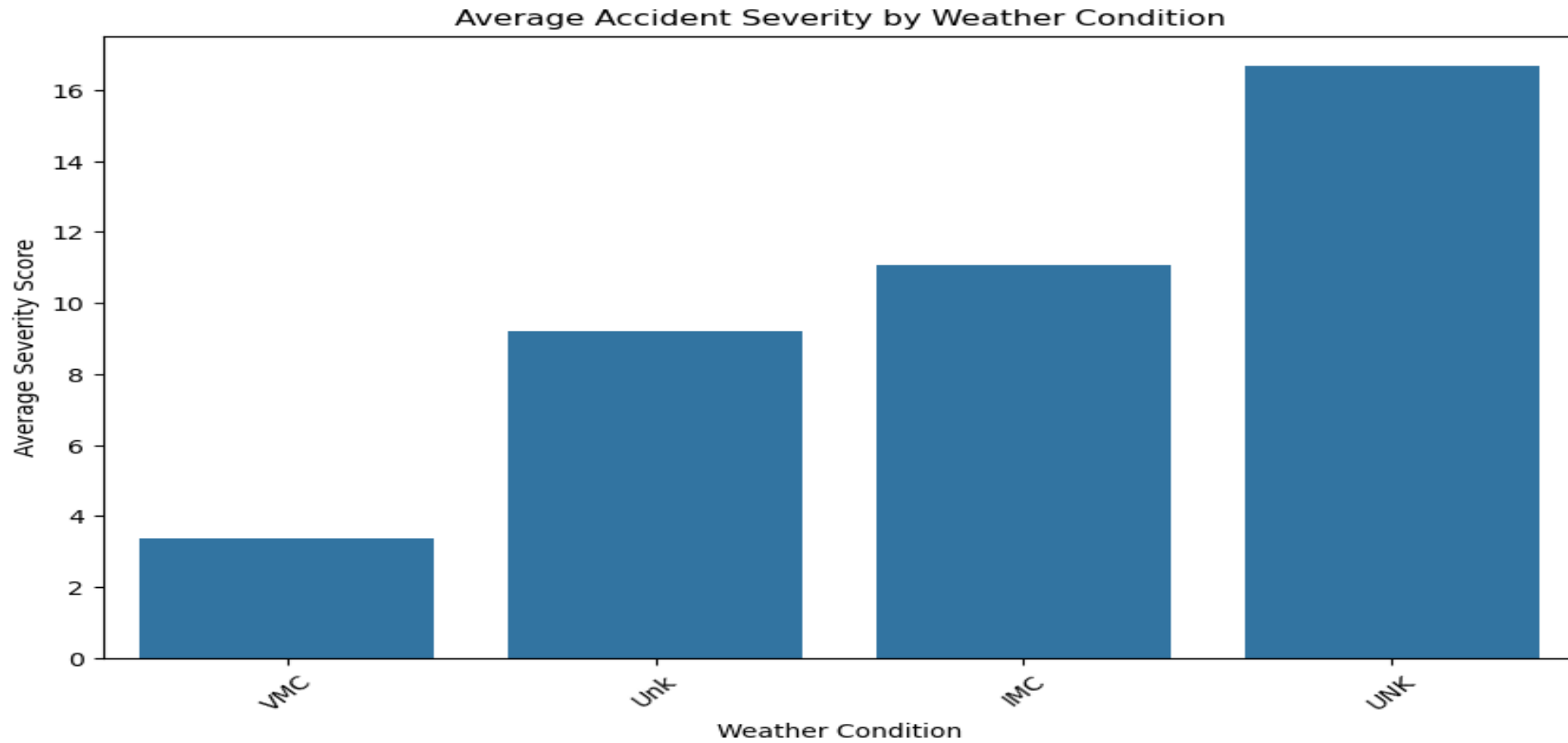
Visualization 2



Finding(viz 2)

- The greater the level of aircraft damage, the higher the number of fatal injuries

Visualization 3



Finding(viz 3)

- Adverse weather conditions are associated with high average severity scores

Overall findings

- Certain aircraft make are deemed safer due to their low average severity score.
- The adverse the weather conditions the high the average severity scores.
- Aircraft damage level is directly proportional to the number of fatal injuries.

Recommendations

- ❖ Certain aircraft makes should be considered for purchase due to their low accident severity score.
- ❖ Operational limits should be put in place during adverse weather conditions.
- ❖ Aircraft models with strong structural resilience should be considered as they are less damaged during accidents, which in turn reduces injury severity.

Future actions

- ❑ Continue monitoring accident data over time
- ❑ Do cost – benefit analysis of the recommended aircraft

THANK
YOU!

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